

## CHAPTER 5: THE LAWS & APPLICATIONS OF EXPONENTS

### Intermediate Practice Worksheet (Level 2)

Designed for Class VII students to transition from textbook exercises to critical thinking problems

## Introduction

This practice worksheet is designed to build directly on **Exercise 5A** and **Exercise 5B**. The difficulty level is calibrated to be **slightly more challenging** than the core textbook exercises. It will help you strengthen your conceptual understanding of exponent laws, scientific notation, real-world applications, and negative indices without entering highly abstract mathematics.

Complete, step-by-step solutions for all 10 problems are provided in the second section of this document.

### Key Concept Reminder

#### Quick Reference of Exponent Laws:

- **Product Law:**  $a^m \times a^n = a^{m+n}$
- **Quotient Law:**  $a^m \div a^n = a^{m-n}$  (for  $m > n$ ) or  $\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$  (for  $m < n$ )
- **Power of a Power Law:**  $(a^m)^n = a^{mn}$
- **Power of a Product:**  $(ab)^n = a^n b^n$       **Power of a Quotient:**  $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- **Negative Exponents:**  $a^{-n} = \frac{1}{a^n}$  and  $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$
- **Zero Exponent:**  $a^0 = 1$  (for any non-zero  $a$ )

## 1 Practice Problems

### Problem 1: Rational Base and Laws Combination

Evaluate and simplify the following expression to its simplest terms:

$$\left[ \left( \frac{2}{3} \right)^2 \right]^3 \times \left( \frac{3}{2} \right)^4 \div \left( \frac{2}{3} \right)^2$$

### Problem 2: Solving Exponential Equations

Solve for the integer value of  $x$ :

$$5^{2x-1} \times 25 = 125^3$$

### Problem 3: Comparing Massive Powers

Without computing the exact numerical values, determine which number is larger:

$$2^{30} \quad \text{or} \quad 3^{20}$$

Explain your step-by-step reasoning clearly.

### Problem 4: Real-World Application (Bacterial Growth)

A specific strain of bacteria under observation doubles in population every 30 minutes. Suppose there are initially 100 bacteria in the culture.

- (i) Write an exponential expression representing the population after  $t$  hours.
- (ii) Calculate the exact number of bacteria in the culture after 3 hours.

### Problem 5: Simplifying Multi-Variable Algebraic Expressions

Simplify the expression below and write your final answer using only positive exponents:

$$\frac{12a^5b^{-3}}{4a^2b^2} \times (ab^2)^2$$

**Problem 6: Standard Scientific Notation and Astronomy Application**

The speed of light in a vacuum is approximately 300,000,000 m/s (metres per second), and the average distance from the Earth to the Sun is approximately 150,000,000,000 m.

- (i) Express both the speed of light and the distance from Earth to the Sun in standard scientific notation ( $a \times 10^k$ , where  $1 \leq a < 10$ ).
- (ii) Using the formula:

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Calculate how many seconds it takes for sunlight to travel to the Earth. Show your exponential division steps and convert the final result to standard integer notation.

**Problem 7: Negative and Fractional Bases under Signs**

Evaluate the expression:

$$(-1)^{101} \times (-2)^4 \times (-3)^3 \div \left(\frac{1}{2}\right)^{-2}$$

Pay close attention to whether negative bases raised to odd or even powers result in positive or negative numbers.

**Problem 8: Algebraic Reduction of Variable Sums**

Determine the exact value of the following fraction. Explain why the result is independent of the value of  $n$ :

$$\frac{2^{n+3} - 2 \times 2^{n+1}}{2^{n+1}}$$

**Problem 9: Paper Folding Thickness Challenge**

A square sheet of paper is folded in half repeatedly. Each fold doubles the number of layers of paper.

- (i) If the paper is folded in half 8 times, how many layers are formed? Express your answer both as a power of 2 and as a whole number.
- (ii) If the original sheet of paper has a uniform thickness of 0.1 mm, calculate the total thickness of the folded paper stack after the 8th fold in millimetres (mm).

**Problem 10: Zero Exponent and Nested Fractions**

Evaluate and simplify the expression to a single rational fraction:

$$\frac{(3^0 + 4^0 + 5^0) \times (a^2 b^3)^0}{\left(\frac{2}{3}\right)^{-2} + \left(\frac{1}{2}\right)^2}$$

(Note: assume  $a \neq 0$  and  $b \neq 0$ ).

## 2 Step-by-Step Solutions

Each solution is detailed below with references to the specific laws of exponents applied.

### Solution 1: Rational Base and Laws Combination

**Problem:** Evaluate  $\left[\left(\frac{2}{3}\right)^2\right]^3 \times \left(\frac{3}{2}\right)^4 \div \left(\frac{2}{3}\right)^2$

**Step 1: Simplify the Power of a Power term** Using the rule  $(a^m)^n = a^{mn}$ :

$$\left[\left(\frac{2}{3}\right)^2\right]^3 = \left(\frac{2}{3}\right)^{2 \times 3} = \left(\frac{2}{3}\right)^6$$

**Step 2: Rewrite bases to make them identical** Observe that the base  $\frac{3}{2}$  is the reciprocal of  $\frac{2}{3}$ . Using the negative reciprocal exponent rule  $\left(\frac{a}{b}\right)^n = \left(\frac{b}{a}\right)^{-n}$ :

$$\left(\frac{3}{2}\right)^4 = \left(\frac{2}{3}\right)^{-4}$$

**Step 3: Combine using Product and Quotient Laws** Now substitute these back into the original expression:

$$\left(\frac{2}{3}\right)^6 \times \left(\frac{2}{3}\right)^{-4} \div \left(\frac{2}{3}\right)^2$$

Applying  $a^m \times a^n = a^{m+n}$  and  $a^m \div a^n = a^{m-n}$ :

$$= \left(\frac{2}{3}\right)^{6+(-4)-2} = \left(\frac{2}{3}\right)^{6-4-2} = \left(\frac{2}{3}\right)^0$$

Since any non-zero number raised to the power of 0 is 1:

$$\left(\frac{2}{3}\right)^0 = 1$$

**Answer:** 1

### Solution 2: Solving Exponential Equations

**Problem:** Solve  $5^{2x-1} \times 25 = 125^3$  for the integer  $x$ .

**Step 1: Express all bases as powers of the prime base 5**

- $25 = 5^2$
- $125 = 5^3 \implies 125^3 = (5^3)^3 = 5^9$  (using  $(a^m)^n = a^{mn}$ )

**Step 2: Substitute and combine LHS terms** Rewrite the equation with base 5:

$$5^{2x-1} \times 5^2 = 5^9$$

Apply the Product Law ( $a^m \times a^n = a^{m+n}$ ) on the Left-Hand Side (LHS):

$$5^{(2x-1)+2} = 5^9$$

$$5^{2x+1} = 5^9$$

**Step 3: Equate the exponents and solve** Since the bases are identical and non-zero, we equate the indices:

$$2x + 1 = 9$$

$$2x = 9 - 1$$

$$2x = 8 \implies x = 4$$

**Answer:**  $x = 4$

**Solution 3: Comparing Massive Powers**

**Problem:** Determine which is larger:  $2^{30}$  or  $3^{20}$ .

**Method: Find a common exponent to compare the bases** The numbers 30 and 20 are large, making direct computation tedious. However, their exponents have a Greatest Common Divisor (GCD) of 10:

$$\text{GCD}(30, 20) = 10$$

Using the Power of a Power rule in reverse,  $a^{mn} = (a^m)^n$ , rewrite both terms with a common outer power of 10:

$$\bullet \quad 2^{30} = 2^{3 \times 10} = (2^3)^{10}$$

$$\bullet \quad 3^{20} = 3^{2 \times 10} = (3^2)^{10}$$

Now compute the inner values:

$$\bullet \quad 2^3 = 8 \implies 2^{30} = 8^{10}$$

$$\bullet \quad 3^2 = 9 \implies 3^{20} = 9^{10}$$

Since the bases can now be directly compared and the exponent is identical:

$$9 > 8 \implies 9^{10} > 8^{10}$$

Therefore,  $3^{20} > 2^{30}$ .

**Answer:**  $3^{20}$  is larger.

**Solution 4: Real-World Application (Bacterial Growth)**

**Problem:** 100 cells initial population, doubles every 30 minutes.

**Part (i): Write an exponential expression for the population after  $t$  hours** Since there are two 30-minute intervals in every hour, the population doubles 2 times per hour.

- After 1 hour, the number of doublings is  $2 \times 1 = 2$ .
- After  $t$  hours, the number of doublings is  $2t$ .

The population  $P(t)$  is given by multiplying the initial population by 2 for each doubling:

$$P(t) = 100 \times 2^{2t}$$

**Part (ii): Calculate the population after 3 hours** Substitute  $t = 3$  into the formula:

$$P(3) = 100 \times 2^{2 \times 3} = 100 \times 2^6$$

Compute  $2^6$ :

$$2^6 = 64$$

Multiply by the initial population:

$$P(3) = 100 \times 64 = 6400 \text{ bacteria}$$

**Answer:** (i)  $P(t) = 100 \times 2^{2t}$ , (ii) 6400 bacteria

**Solution 5: Simplifying Multi-Variable Algebraic Expressions**

**Problem:** Simplify  $\frac{12a^5b^{-3}}{4a^2b^2} \times (ab^2)^2$  and write with positive exponents.

**Step 1: Simplify the first fraction** Divide coefficients and apply the Quotient Law ( $a^m \div a^n = a^{m-n}$ ):

$$\frac{12a^5b^{-3}}{4a^2b^2} = \left(\frac{12}{4}\right) a^{5-2}b^{-3-2} = 3a^3b^{-5}$$

**Step 2: Expand the squared term** Apply the Power of a Product Law  $(xy)^n = x^ny^n$  and  $(x^m)^n = x^{mn}$ :

$$(ab^2)^2 = a^2(b^2)^2 = a^2b^4$$

**Step 3: Multiply the simplified terms** Now multiply the two intermediate results using the Product Law ( $a^m \times a^n = a^{m+n}$ ):

$$3a^3b^{-5} \times a^2b^4 = 3a^{3+2}b^{-5+4} = 3a^5b^{-1}$$

**Step 4: Express with positive exponents** Using the rule  $b^{-1} = \frac{1}{b}$ :

$$3a^5b^{-1} = \frac{3a^5}{b}$$

**Answer:**  $\frac{3a^5}{b}$

**Solution 6: Standard Scientific Notation and Astronomy Application**

**Problem:** Speed = 300,000,000 m/s, Distance = 150,000,000,000 m.

**Part (i): Convert to scientific notation**

- **Speed of light:**  $300,000,000 = 3 \times 10^8$  m/s (decimal moved 8 places left)
- **Distance to Sun:**  $150,000,000,000 = 1.5 \times 10^{11}$  m (decimal moved 11 places left)

**Part (ii): Calculate travel time** Apply the formula:

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{1.5 \times 10^{11}}{3 \times 10^8}$$

Separate the decimals and exponents:

$$\text{Time} = \left(\frac{1.5}{3}\right) \times \frac{10^{11}}{10^8}$$

Calculate each part:

$$\frac{1.5}{3} = 0.5$$

Applying the Quotient Law:

$$\frac{10^{11}}{10^8} = 10^{11-8} = 10^3$$

Combine and simplify to standard integer notation:

$$\text{Time} = 0.5 \times 10^3 = 0.5 \times 1000 = 500 \text{ seconds}$$

**Answer:** (i) Speed =  $3 \times 10^8$  m/s, Distance =  $1.5 \times 10^{11}$  m, (ii) 500 seconds

**Solution 7: Negative and Fractional Bases under Signs****Problem:** Evaluate  $(-1)^{101} \times (-2)^4 \times (-3)^3 \div \left(\frac{1}{2}\right)^{-2}$ **Step 1: Evaluate each term individually, noting the signs**

- $(-1)^{101}$ : Since the exponent 101 is odd, the result is negative:

$$(-1)^{101} = -1$$

- $(-2)^4$ : Since the exponent 4 is even, the result is positive:

$$(-2)^4 = (-2) \times (-2) \times (-2) \times (-2) = 16$$

- $(-3)^3$ : Since the exponent 3 is odd, the result is negative:

$$(-3)^3 = (-3) \times (-3) \times (-3) = -27$$

- $\left(\frac{1}{2}\right)^{-2}$ : Apply the negative exponent reciprocal rule  $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$ :

$$\left(\frac{1}{2}\right)^{-2} = \left(\frac{2}{1}\right)^2 = 2^2 = 4$$

**Step 2: Perform the operations in sequence** Substitute the evaluated terms back:

$$(-1) \times 16 \times (-27) \div 4$$

Multiply the first two terms:

$$(-1) \times 16 = -16$$

Multiply by the third term:

$$(-16) \times (-27) = 432 \quad (\text{negative} \times \text{negative} = \text{positive})$$

Divide by the final term:

$$432 \div 4 = 108$$

**Answer:** 108**Solution 8: Algebraic Reduction of Variable Sums****Problem:** Show that  $\frac{2^{n+3} - 2 \times 2^{n+1}}{2^{n+1}}$  is independent of  $n$  and find its value.**Step 1: Simplify terms in the numerator using exponent laws** Using the Product Law in reverse,  $a^{m+n} = a^m \times a^n$ :

$$2^{n+3} = 2^{n+1} \times 2^2 = 4 \times 2^{n+1}$$

**Step 2: Factor out the common term  $2^{n+1}$  in the numerator** Rewrite the numerator with this substitution:

$$\text{Numerator} = 2^{n+3} - 2 \times 2^{n+1} = 4 \times 2^{n+1} - 2 \times 2^{n+1}$$

Factoring out  $2^{n+1}$ :

$$= (4 - 2) \times 2^{n+1} = 2 \times 2^{n+1} = 2^{n+2}$$



**Step 3: Simplify the fraction** Substitute the simplified numerator back into the fraction:

$$\frac{2 \times 2^{n+1}}{2^{n+1}}$$

Since  $2^{n+1}$  appears in both the numerator and the denominator, they cancel out completely:

$$= 2$$

Since the variable  $n$  cancels out completely during simplification, the value of the expression is always 2, regardless of the value of  $n$ .

**Answer:** 2

**Solution 9: Paper Folding Thickness Challenge**

**Problem:** Repeated paper folding starting with 1 layer of 0.1 mm thickness.

**Part (i): Find layers formed after 8 folds** The number of layers doubles with each fold.

- 1 fold  $\Rightarrow 2^1 = 2$  layers.
- 2 folds  $\Rightarrow 2^2 = 4$  layers.
- 8 folds  $\Rightarrow 2^8$  layers.

Calculate  $2^8$ :

$$2^8 = 256$$

So,  $2^8 = 256$  layers are formed.

**Part (ii): Calculate total thickness after 8 folds** The total thickness is the number of layers multiplied by the thickness of a single layer:

$$\text{Total Thickness} = \text{Number of Layers} \times \text{Thickness of 1 Layer}$$

$$\text{Total Thickness} = 256 \times 0.1 \text{ mm} = 25.6 \text{ mm}$$

(Note: 25.6 mm is equivalent to 2.56 cm, which is remarkably thick for just folding paper!)

**Answer:** (i)  $2^8 = 256$  layers, (ii) 25.6 mm

**Solution 10: Zero Exponent and Nested Fractions**

**Problem:** Evaluate  $\frac{(3^0+4^0+5^0) \times (a^2b^3)^0}{\left(\frac{2}{3}\right)^{-2} + \left(\frac{1}{2}\right)^2}$

**Step 1: Simplify the numerator** Using the zero exponent rule  $x^0 = 1$  (for  $x \neq 0$ ):

- $3^0 = 1, 4^0 = 1, 5^0 = 1 \Rightarrow (3^0 + 4^0 + 5^0) = 1 + 1 + 1 = 3$
- $(a^2b^3)^0 = 1$  (since  $a \neq 0, b \neq 0$ )

Multiply these results together:

$$\text{Numerator} = 3 \times 1 = 3$$

**Step 2: Simplify the denominator**

- Apply the reciprocal rule  $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$  to the first term:

$$\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^2 = \frac{3^2}{2^2} = \frac{9}{4}$$

- Evaluate the second term:

$$\left(\frac{1}{2}\right)^2 = \frac{1^2}{2^2} = \frac{1}{4}$$

- Sum the terms:

$$\text{Denominator} = \frac{9}{4} + \frac{1}{4} = \frac{10}{4} = \frac{5}{2}$$

**Step 3: Divide the simplified numerator by the simplified denominator**

$$\frac{\text{Numerator}}{\text{Denominator}} = \frac{3}{\left(\frac{5}{2}\right)} = 3 \times \frac{2}{5} = \frac{6}{5}$$

Converting to decimal form:

$$\frac{6}{5} = 1.2$$

**Answer:**  $\frac{6}{5}$  (or 1.2)